1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, TLS/local lymphoid organization, systemic mutational/repair state (MSH6/MSI/bTMB) and adaptive immune checkpoints (PD-L1) all influence NMIBC clinical behavior — but it is unclear why some TLS-rich tumors still fail to benefit from intravesical BCG. 2) Tier 1 paths used: Core Path 2 (bTMB ↔ BCG ↔ PD-L1) and Core Path 3 (MSH6 as prognostic/MMR axis) form a mechanistic backbone linking mismatch-repair status and adaptive PD-L1-mediated immune suppression to BCG responsiveness; I chose these because they directly explain an immune evasion mechanism that can blunt local lymphoid advantages. 3) Tier 2 evidence: ADNP-in-CTCs (Core Path 1) motivates a liquid-biopsy surveillance angle but is mechanistically orthogonal to the immediate immune-evasion question; MSI/dMMR, TP53 and ER status from Tier 2 provide supportive context for genomic instability and immune visibility that can modulate MSH6 effects and PD-L1 induction. 4) Endpoint choice: I selected therapy_response (BCG response proxy) because the most actionable clinical question for NMIBC is which patients benefit from intravesical immunotherapy versus who needs escalation to systemic/combination immunotherapy; therapy response is also directly linked to the immune-mechanistic paths. 5) Deliberate exclusions: I excluded ADNP from the core hypothesis to keep the mechanistic focus tight (immune evasion → therapy failure) and relegated it to secondary_modifiers as a surveillance biomarker; I also excluded bTMB as a core anchor (kept it as a modifier) to limit anchors to MSH6 and PD-L1 which give a clear MMR → adaptive checkpoint mechanism. I deliberately did not expand the hypothesis with TP53/ER/other exploratory paths to preserve the one-imaging/≤2-gene constraint and maximize mechanistic clarity?

Working Hypothesis

TLS-like lymphoid aggregates in NMIBC generally reflect organized local anti-tumor immunity, but when tumors harbor MSH6 deficiency, this TLS presence may not translate into BCG responsiveness because MSH6 loss drives MMR deficiency and hypermutation that upregulates PD-L1 and promotes T cell exhaustion, decoupling TLS from effective intravesical immunotherapy. Therefore, NMIBC tumors with TLS and MSH6 deficiency plus high PD-L1 expression will show reduced therapy_response to BCG, identifying patients who may require PD-1/PD-L1–directed or alternative systemic or combination immunotherapies rather than standard intravesical treatment.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSH6, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 as prognostic/MMR axis) form a mechanistic backbone linking mismatch-repair status and adaptive PD-L1-mediated immune suppression to BCG responsiveness; I chose these because they directly explain an immune evasion mechanism that can blunt local lymphoid advantages.

PD-L1: PD-L1) and Core Path 3 (MSH6 as prognostic/MMR axis) form a mechanistic backbone linking mismatch-repair status and adaptive PD-L1-mediated immune suppression to BCG responsiveness; I chose these because they directly explain an immune evasion mechanism that can blunt local lymphoid advantages.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	ADNP	Supporting modifier from knowledge graph
Secondary	bTMB	Supporting modifier from knowledge graph
Secondary	MSI/dMMR	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	ER	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: assemble NMIBC cohorts treated with BCG with available tissue;

Prediction 2: assess TLS density via multiplex IHC and characterize TLS maturation;

Prediction 3: determine MSH6 status (IHC/genomic) and PD-L1 (CD274) expression;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- ADNP: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.